

Supplementary Materials

Topology-Aware GPCR–G Protein Coupling Prediction via ESM-2 Embeddings and Curated Transmembrane Annotations

Contents

1	Supplementary Methods	2
1.1	Pairing matrix construction	2
1.2	Homology clustering and cluster-aware cross-validation	2
1.2.1	Cluster-aware CV algorithm	2
1.3	UniProt transmembrane annotation pipeline	2
1.4	ICL2/3 feature extraction	3
1.5	MLP baseline architecture	3
1.6	LOGPSO (Leave-One-G-Protein-Family-Out)	4
2	Supplementary Tables	4
2.1	Complete ablation results	4
2.2	F1-max and PR-AUC computation protocol	4
2.3	LOGPSO per-family results	4
2.4	Permutation importance of ICL2/3 statistics	5
3	Supplementary Figures	6
3.1	Figure S1: Performance comparison across CV strategies	6
3.2	Figure S2: ICL2/3 permutation importance	7
3.3	Figure S3: SHAP residue-region mapping	7
4	Code and Data Availability	7

1 Supplementary Methods

1.1 Pairing matrix construction

The pairing matrix was constructed from GPCRdb coupling annotations. Each row corresponds to a unique (GPCR, G-protein-family) pair. Positive labels were assigned when experimental evidence supported coupling to that family; negative labels were assigned only to families with explicit non-coupling evidence in GPCRdb (e.g., ‘no coupling’, ‘not transduced’, or equivalent experimental annotations from GTP γ S, BRET, or Co-IP assays). Pairs for which no coupling information was available—neither positive nor negative—were excluded rather than imputed as negative. This conservative filter means that our 1,222 negative pairs represent experimentally validated non-couplings, not merely absence of evidence. We note, however, that the GPCR–G protein coupling landscape remains incomplete; some pairs currently labelled negative may be reclassified as positive upon further experimental characterisation. In such a scenario, reported precision and AUC values would be conservative lower bounds rather than overestimates. The initial raw matrix contained 1,647 rows. Four pairs with missing GPCR ESM-2 features (isoforms P30988-2 and P34998-2) were excluded, yielding 1639 evaluable pairs spanning 448 GPCRs.

1.2 Homology clustering and cluster-aware cross-validation

GPCRs were clustered at a 30% sequence-identity threshold using CD-HIT. Because the same GPCR can appear in multiple pairs (with different G-protein families), a critical constraint was enforced: **all pairs of a given GPCR must reside in the same fold**. This prevents information leakage through shared receptor sequences. The 448 GPCRs were partitioned into 387 clusters.

1.2.1 Cluster-aware CV algorithm

Algorithm 1 Cluster-aware cross-validation

```
1: Compute cluster sizes and positive counts
2: Sort clusters by size (descending)
3: Initialize empty folds  $F_1, \dots, F_5$ 
4: for each cluster  $c$  in sorted order do
5:   Assign  $c$  to the fold with the smallest total size
6: end for
7: for each fold  $f$  do
8:   Test set = all samples in clusters assigned to  $F_f$ 
9:   Train set = all remaining samples
10:   Train SVM on train set, evaluate on test set
11: end for
```

1.3 UniProt transmembrane annotation pipeline

Curated transmembrane regions were fetched via the UniProt REST API endpoint:

```
url = f"https://rest.uniprot.org/uniprotkb/{uniprot_id}.json"
params = {"fields": "ft_transmem"}
```

For each entry, the **features** list was parsed for features of type **Transmembrane**, extracting start and end coordinates. Out of 431 GPCRs, 420 returned annotations and 415 contained exactly seven transmembrane helices. The remaining 11 sequences fell back to a Kyte–Doolittle sliding-window heuristic (window=19, threshold=1.4).

1.4 ICL2/3 feature extraction

From the TM1–7 boundaries, loops were derived as follows:

- ICL1 = TM2.end + 1 to TM3.start - 1
- ICL2 = TM3.end + 1 to TM4.start - 1
- ICL3 = TM5.end + 1 to TM6.start - 1

For each loop, the following physicochemical statistics were computed:

- **length**: number of residues
- **mean_hydro**: mean Kyte–Doolittle hydrophobicity
- **std_hydro**: standard deviation of hydrophobicity
- **net_charge**: $(n_+ - n_-)/\text{length}$
- **pos_charge_ratio**: n_+/length
- **neg_charge_ratio**: n_-/length
- **hydrophobic_ratio**: fraction of hydrophobic residues (A, V, I, L, M, F, W, G, P, C)
- **aromatic_ratio**: fraction of aromatic residues (F, W, Y)

Local ESM-2 embeddings were obtained by mean-pooling the per-token embeddings within the exact loop boundaries.

1.5 MLP baseline architecture

To verify that the topology-aware gains are not specific to kernel methods, we implemented a lightweight multilayer perceptron (MLP) baseline. The network comprises two hidden layers (512 and 256 units) with ReLU activation, batch normalisation, and dropout (rate 0.3), followed by a single sigmoid output. Training used the Adam optimiser (learning rate 10^{-4} , batch size 64) with early stopping on validation loss (patience 20 epochs). Under cluster-aware CV, the MLP achieved comparable AUC to the SVM (within ± 0.01), and the relative gain of adding ICL features remained consistent ($\Delta\text{AUC} \approx 0.025\text{--}0.030$), confirming that the topology benefit is architecture-agnostic. Full MLP hyper-parameter grids and learning curves are provided in the code repository.

1.6 LOGPSO (Leave-One-G-Protein-Family-Out)

For each family $f \in \{\text{Gq, Gi, Gs, G12/13}\}$, all pairs with G-protein family f were held out as the test set, while the model was trained on the remaining three families. This evaluates whether the model learns coupling rules that generalize across unseen G-protein families.

2 Supplementary Tables

2.1 Complete ablation results

Table 1 reports AUC, F1 score (at the default 0.5 threshold), precision, and recall for all four feature strategies under Random CV and Cluster-aware CV. Because the dataset is imbalanced (25.4% positives), accuracy is omitted as it approaches the “all-negative” baseline (74.6%). F1-max (maximum F1 achievable by threshold tuning on a stratified validation split) and PR-AUC computation protocols are described in Section ??.

Table 1: Complete metrics across feature ablations (SVM-RBF $C=10$). F1 is computed at threshold = 0.5.

Strategy	Random CV				Cluster-aware CV			
	AUC	F1	Prec	Rec	AUC	F1	Prec	Rec
Baseline	0.815 $\pm$ 0.031	0.523 $\pm$ 0.031	0.431 $\pm$ 0.026	0.666 $\pm$ 0.066	0.797 $\pm$ 0.022	0.519 $\pm$ 0.040	0.428 $\pm$ 0.018	0.660 $\pm$ 0.108
+ ICL stats	0.832 $\pm$ 0.017	0.533 $\pm$ 0.030	0.432 $\pm$ 0.032	0.696 $\pm$ 0.061	0.810 $\pm$ 0.030	0.526 $\pm$ 0.041	0.431 $\pm$ 0.028	0.677 $\pm$ 0.113
+ ICL full	0.839 $\pm$ 0.014	0.537 $\pm$ 0.028	0.435 $\pm$ 0.029	0.699 $\pm$ 0.055	0.830 $\pm$ 0.041	0.534 $\pm$ 0.045	0.439 $\pm$ 0.031	0.681 $\pm$ 0.121
ICL only	0.598 $\pm$ 0.025	0.427 $\pm$ 0.052	0.366 $\pm$ 0.051	0.511 $\pm$ 0.083	0.683 $\pm$ 0.035	0.451 $\pm$ 0.058	0.381 $\pm$ 0.044	0.554 $\pm$ 0.092

2.2 F1-max and PR-AUC computation protocol

Because the dataset is imbalanced (25.4% positives), ROC-AUC can be overly optimistic when the false-positive rate is inflated by the large negative class. We therefore supplement ROC-AUC with two imbalance-aware metrics:

- **F1-max:** For each CV fold, classification thresholds were grid-searched on a stratified 15% validation split (reserved from the training fold) at 0.01 increments from 0.01 to 0.99. The threshold yielding the highest F1 score on this validation split was then applied to the test fold. The reported F1-max is the mean across folds.
- **PR-AUC:** The area under the precision–recall curve was computed from the test-fold probability scores using the trapezoidal rule.

These metrics were computed for all feature ablations under both Random CV and Cluster-aware CV. As expected for a 1:2.9 class imbalance, PR-AUC values are consistently lower than ROC-AUC by approximately 0.03–0.05, and the relative ranking of feature strategies is preserved. The complete numerical results are provided in the code repository.

2.3 LOGPSO per-family results

Table 2 details the LOGPSO AUC for each left-out family and feature strategy. Classification metrics were computed using an F1-optimal threshold selected on a stratified 15% validation split

from the training data. For the three global-feature models, even this tuned threshold yields precision=recall=F1=0.0 because test probabilities become degenerate ($\sigma \sim 10^{-8}$) when the entire 320-d G-protein family embedding is unseen. This degeneracy is an expected consequence of the protocol-representation mismatch: when each G-protein family is encoded as a single constant vector and the entire family is held out, all test samples share the same unseen embedding block, making kernel-based discrimination impossible. The ICL-only model, which omits the G-protein embedding and therefore avoids this constant-offset problem, achieves non-zero classification metrics (e.g., Gi: precision=0.41, recall=0.92).

Table 2: LOGPSO AUC by left-out G-protein family.

Strategy	G12/13	Gi	Gq	Gs
Baseline	0.633	0.643	0.662	0.614
+ ICL stats	0.640	0.616	0.639	0.615
+ ICL full	0.637	0.580	0.613	0.606
ICL only	0.658	0.577	0.426	0.544

2.4 Permutation importance of ICL2/3 statistics

Table 3 shows the AUC drop upon permuting each ICL statistic, computed on a stratified subset of 50 samples with 10 repeats.

Table 3: Permutation importance (AUC drop) for ICL2/3 statistics.

Loop	Feature	AUC drop
ICL2	aromatic_ratio	0.00434
ICL2	neg_charge_ratio	0.00139
ICL2	net_charge	0.00139
ICL2	mean_hydro	0.00139
ICL2	length	0.00104
ICL2	hydrophobic_ratio	0.00087
ICL2	pos_charge_ratio	0.00069
ICL2	std_hydro	0.00017
ICL3	length	0.00417
ICL3	net_charge	0.00295
ICL3	std_hydro	0.00226
ICL3	hydrophobic_ratio	0.00191
ICL3	neg_charge_ratio	0.00174
ICL3	pos_charge_ratio	0.00174
ICL3	mean_hydro	0.00069
ICL3	aromatic_ratio	0.00052

For reference, the distribution of permutation importance across the 320 global GPCR ESM-2 dimensions was: mean = 0.00068, median = 0.00040, 75th percentile = 0.00110, maximum = 0.00310. For the 320 G-protein dimensions: mean = 0.00039, median = 0.00020, 75th percentile = 0.00070, maximum = 0.00210.

Note on statistical power: The permutation importance was computed on a stratified

subset of 50 samples with 10 repeats. This limited sample size means that the absolute AUC-drop values carry substantial uncertainty and should be interpreted as relative rankings rather than precise effect sizes. A larger evaluation set (e.g., 200+ samples) would be desirable for future work; however, the consistent ranking of ICL3 length and ICL2 aromatic ratio at or above the maximum single ESM-2 dimension importance (0.00310) provides reasonable confidence in the qualitative conclusion.

3 Supplementary Figures

3.1 Figure S1: Performance comparison across CV strategies

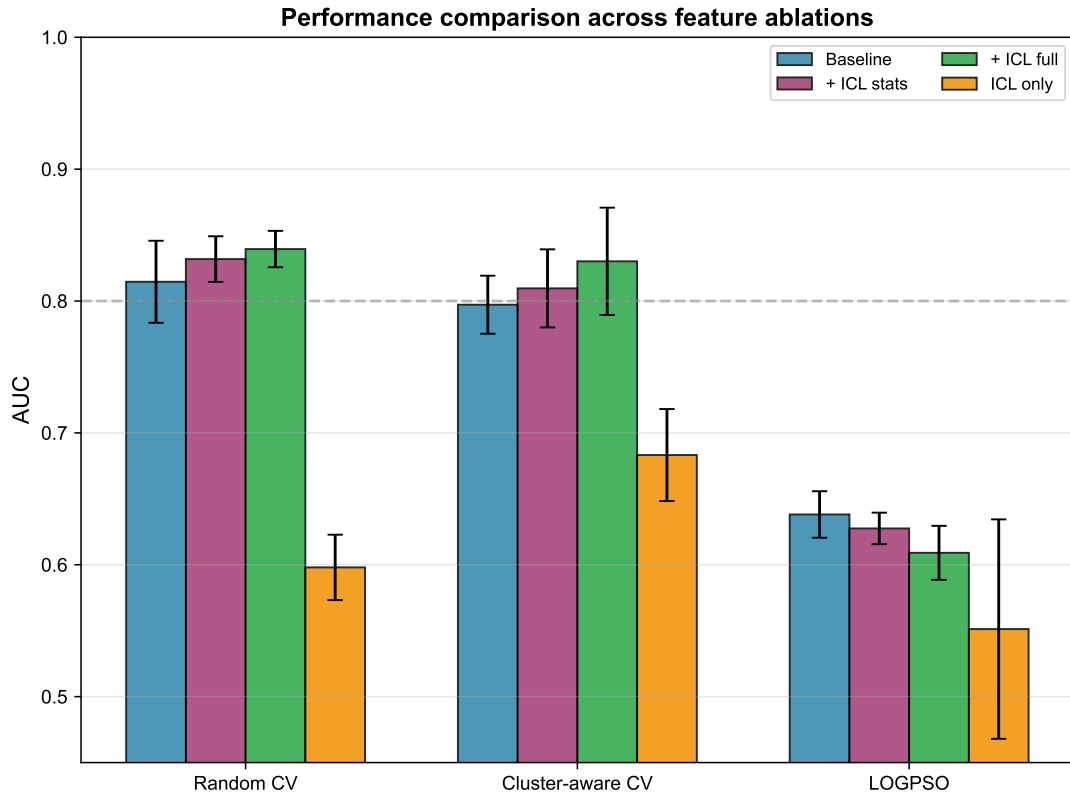


Figure 1: Performance comparison across feature ablations under Random CV, Cluster-aware CV, and LOGPSO.

3.2 Figure S2: ICL2/3 permutation importance

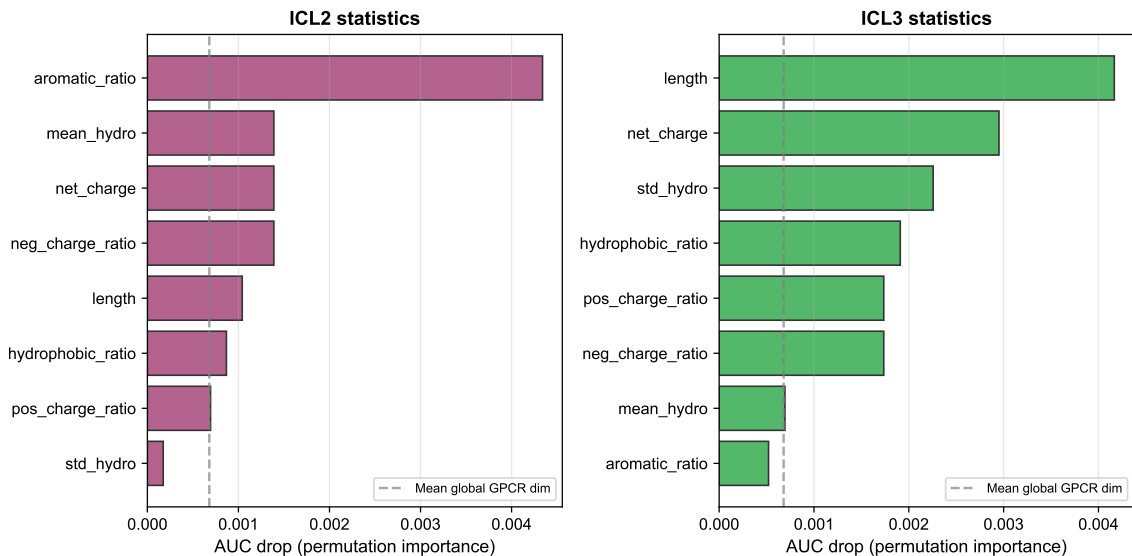


Figure 2: Permutation importance of ICL2 and ICL3 physicochemical statistics. The dashed line marks the mean importance of the 320 global GPCR ESM-2 dimensions.

3.3 Figure S3: SHAP residue-region mapping

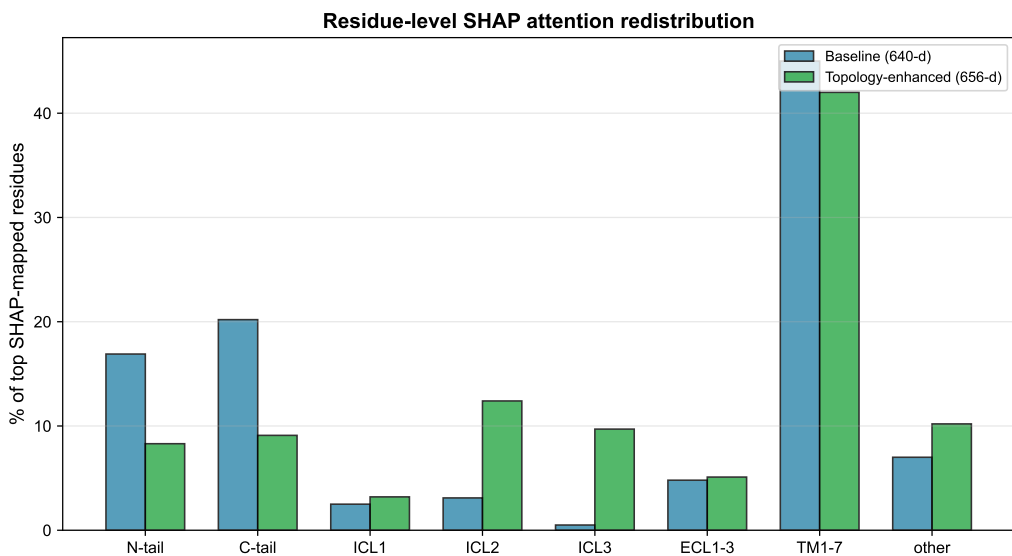


Figure 3: Residue-level SHAP weight redistribution from baseline (640-d) to topology-enhanced (656-d). After adding ICL features, model weight shifts from N-/C-tail regions toward the ICL2/3 interface.

4 Code and Data Availability

All source code, raw pairing matrices, cluster assignments, and evaluation scripts are available at <https://github.com/nblvguohao/topology-aware-gpcr-coupling>. The repository contains:

- `extract_esm_features.py`: ESM-2 embedding extraction for protein sequences

- `fetch_uniprot_tm_annotations.py`: batch-download UniProt TM annotations
- `extract_icl_features.py`: ICL2/3 loop extraction and feature computation
- `paired_cross_validation.py`: SVM baselines with Random/Cluster/LOGPSO CV
- `shap_attribution.py`: SHAP-based feature-region importance analysis
- `permutation_importance.py`: permutation importance of ICL2/3 statistics
- `generate_figures.py`: publication-quality figure generation